

$$x = ky$$

$$12 = \cancel{k} \frac{1}{8} \times y$$

$$y = 12 \times 8$$

$$= 96$$

$$\frac{x}{y} = k$$

proportion

∞

$$x \propto y$$

$$x = ky$$

$$1 = k \times 8$$

$$k = \frac{1}{8}$$

$$x = ky$$

$$x = \frac{1}{8} \times 56$$

$$= 7$$

$$\begin{aligned}
 m &\propto n \\
 m &= k \cdot n \\
 k &= \frac{m}{n}
 \end{aligned}$$

$$m \propto n$$

$$3 \propto 12$$

$$3 = k \times 12$$

$$k = \frac{3}{12}$$

$$k = \frac{1}{4}$$

$$m = k \times n$$

$$6.5 = \frac{1}{4} \times n$$

$$n = 6.5 \times 4$$

$$= 26$$

$$k = \frac{m}{n}$$

$$\frac{1}{4} \neq \frac{6.5}{n}$$

$$n = 6.5 \times 4$$

$$\begin{aligned}
 \textcircled{5} \quad y &\propto \sqrt{x} \\
 y &= k\sqrt{x} \\
 24 &= k\sqrt{16} \\
 24 &= k \times 4 \\
 k &= \frac{24}{4} \\
 &= \underline{6} \\
 y &= 6\sqrt{x}
 \end{aligned}$$

$$\begin{aligned}
 \text{Let lab} &= x \\
 \text{Revenue} &= y
 \end{aligned}$$

$$\begin{aligned}
 x &= 4 \\
 y &= 1000 \text{ ₹}
 \end{aligned}$$

$$\begin{aligned}
 x &= 17 \\
 y &= ?
 \end{aligned}$$

$$\boxed{x \propto y}$$

$$x = ky$$

D. V.

$$\boxed{k = \frac{x}{y}}$$

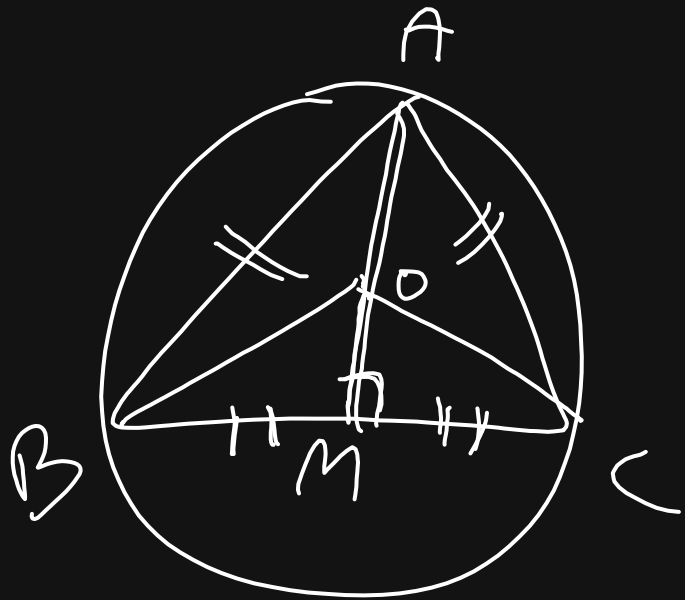
$$x \propto \frac{1}{y}$$

$$x = k \times \frac{1}{y}$$

$$\Rightarrow k = x \times y$$

I. V.

$$\begin{aligned} 4 &= 2 \\ 8 &= 1 \end{aligned}$$



$$BM = CM$$

Altitudes = median

I so
(P. B)

$$IT \quad 100500 - 125500 = \boxed{1}$$

OT

$$\frac{1500}{2}$$

$$750$$

$$+ 750$$

8000
6500

$$180\%$$

2

⑤ $x \propto \frac{1}{\sqrt{y}}$

$$x = k \times \frac{1}{\sqrt{y}}$$

$$\begin{aligned} k &= x \times \sqrt{y} \\ &= 40 \times \sqrt{16} \\ &= 40 \times 4 \\ &= 160 \end{aligned}$$

$$x = 10 \times \frac{1}{\sqrt{y}}$$

$$10 = 160 \times \frac{1}{\sqrt{y}}$$

$$\sqrt{y} = \frac{160}{10}$$

$$\sqrt{y} = 16$$

$$y = (16)^2$$

$$= 256$$

sq. both sides

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\frac{-8 \pm \sqrt{256}}{2 \times 1}$$

$$2 \times 1$$

$$\frac{-8 + 16}{2} \text{ OR } \frac{-8 - 16}{2}$$

$$\begin{array}{r} 84 \\ \underline{8} \end{array}$$

$$4$$

$$\begin{array}{r} -24^{12} \\ \underline{8} \end{array}$$

$$-12$$

$$x^2 + 8x - 48$$

$$a = 1$$

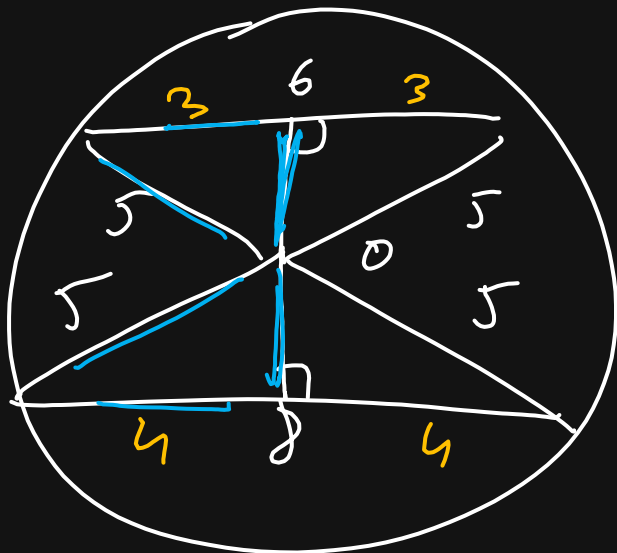
$$b = 8$$

$$c = -48$$

$$b^2 - 4ac = 8^2 - 4 \times 1 \times (-48)$$

$$= 64 + 192$$

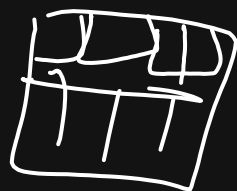
$$= \boxed{256}$$



Q.3]

I ✓

Let apples be x
bananas be y



$$x \propto \frac{1}{y}$$

$$x = k \times \frac{1}{y}$$

(38)

$$x = 24$$

$$y = 27$$

$$x = 36$$

$$y = ?$$